

DEVOPS LAB 01

GitHub Identity, Student Portal Registration,
and Ubuntu Server Installation

Name: Fatima Noor Ul Imran

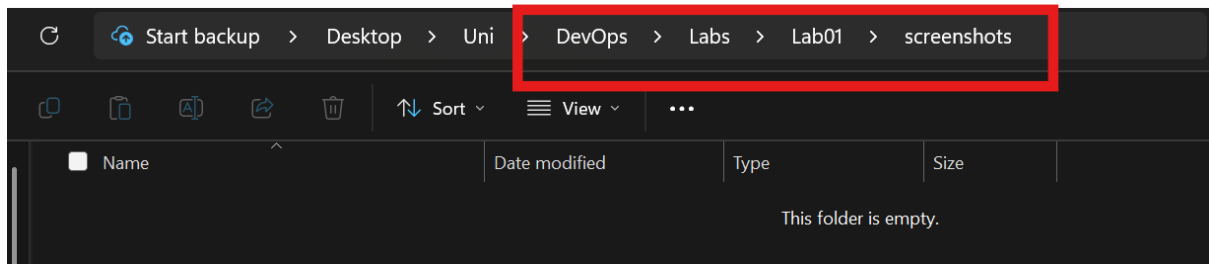
Registration Number: 2023-BSCS-33

Course Code: BCS-649

Section:A

GitHub Username: fatimanoorulimran

Getting Started

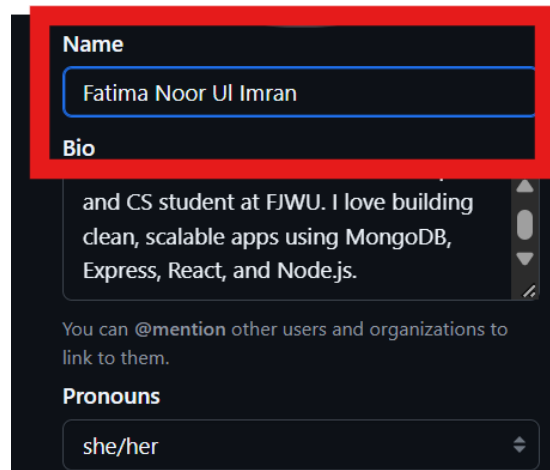


Folder structure Labs/Lab01/screenshots created inside the DevOps repository.

Task 1: Create and Configure Your GitHub Account

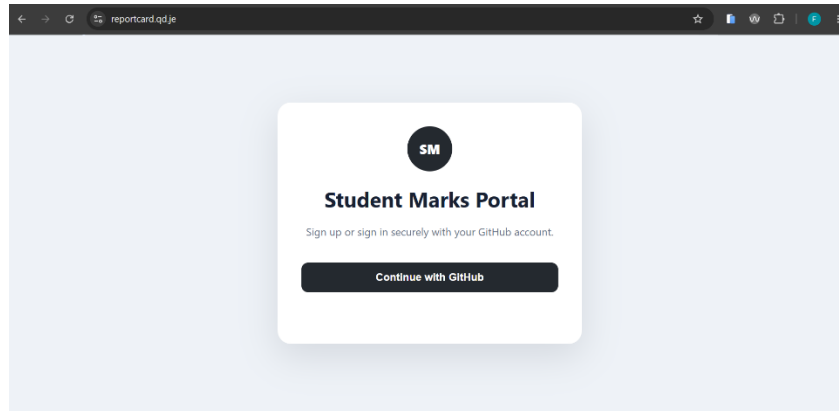


GitHub profile showing my full name and username fatimanoorulimran.

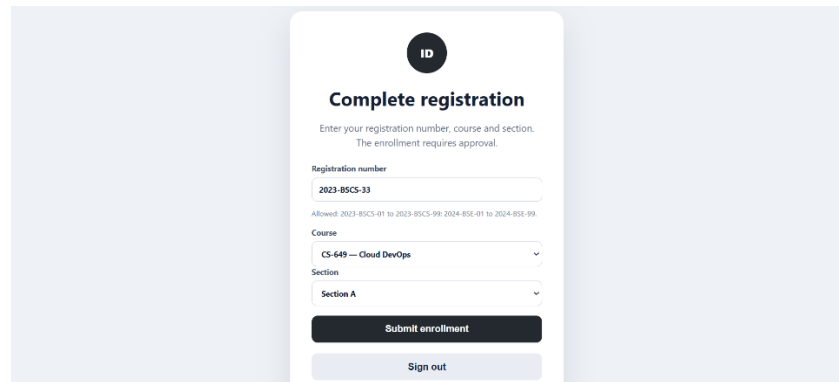


Public profile with my complete actual name in the Name field.

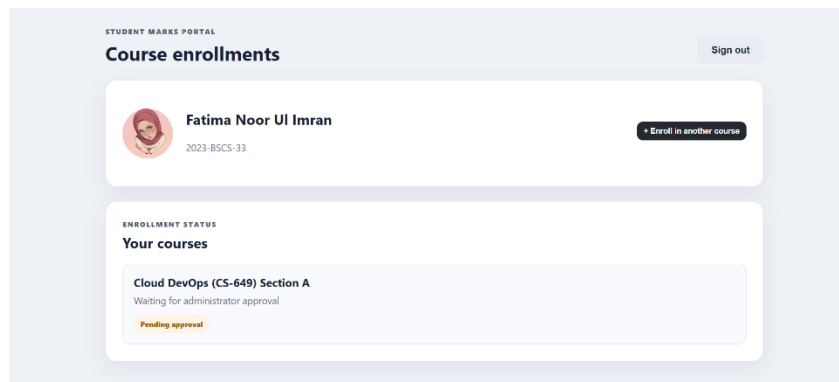
Task 2: Register in the Student Marks Portal



Student Marks Portal login page.

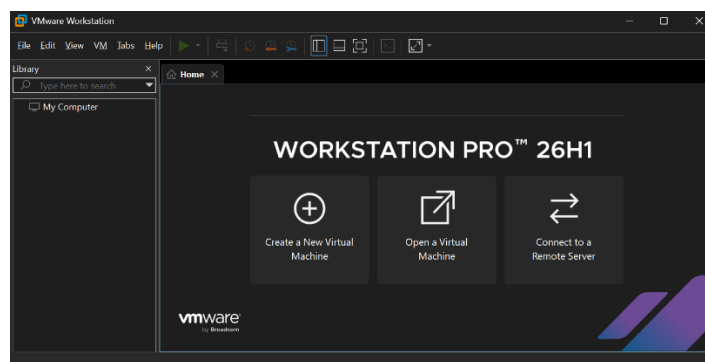


Portal Registration Form after Continuing with GitHub.

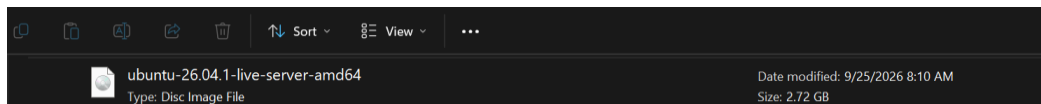


Submitted enrollment showing pending approval status.

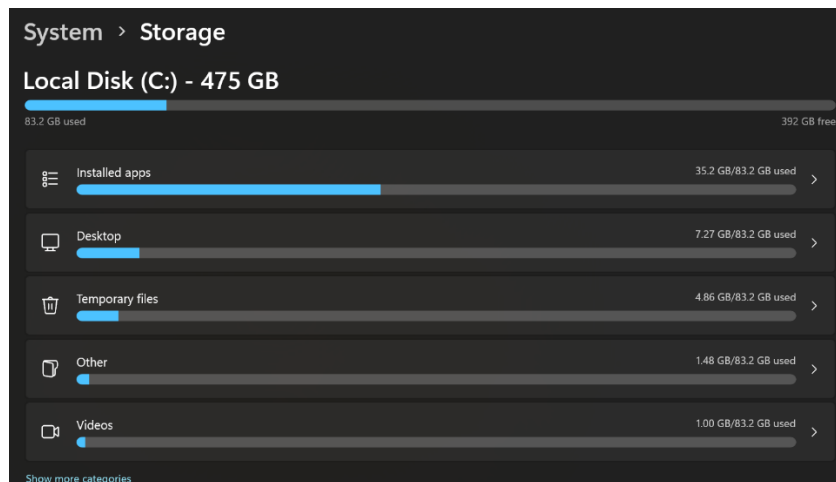
Task 3: Download VMware Workstation Pro and Ubuntu Server



VMware Workstation Pro installed and running.

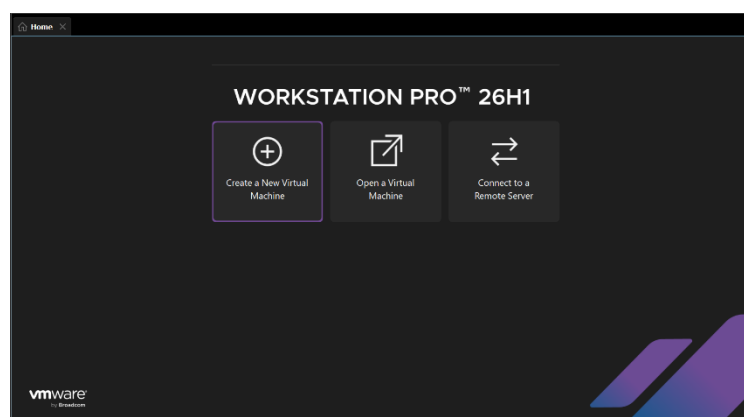


Downloaded Ubuntu Server LTS ISO with filename visible.

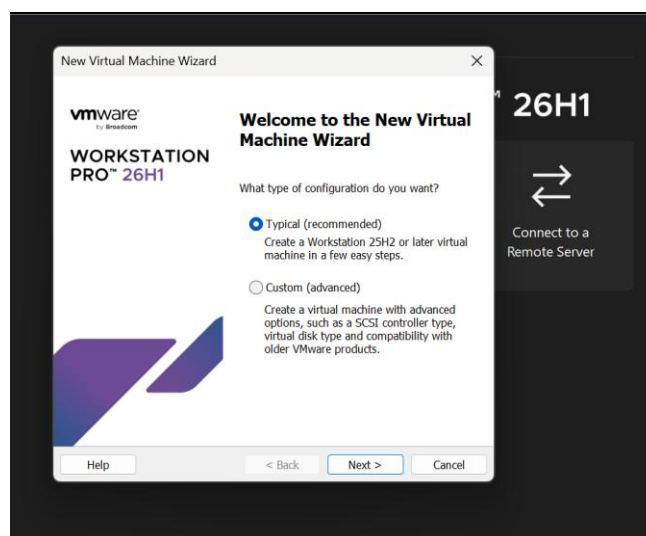


Available storage confirming at least 20 GB free.

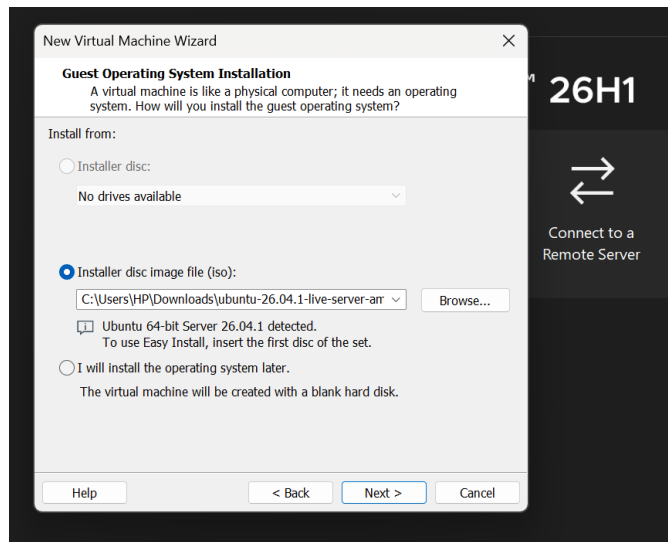
Task 4: Create the Ubuntu Server Virtual Machine



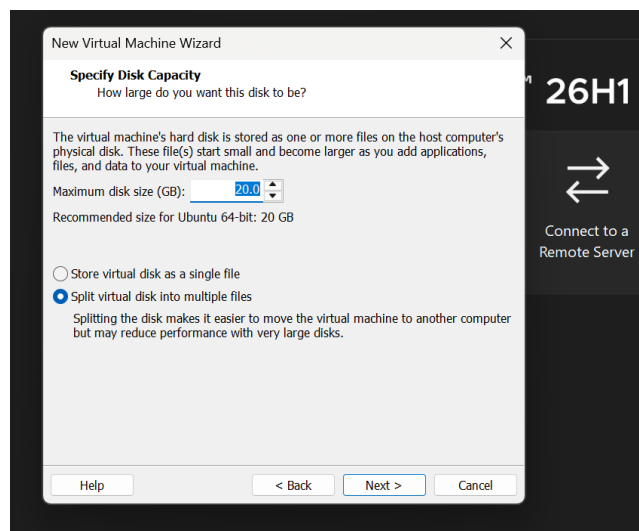
New Virtual Machine Wizard in VMware.



Typical (recommended) configuration selected.



Ubuntu Server ISO selected as the installer disc image.

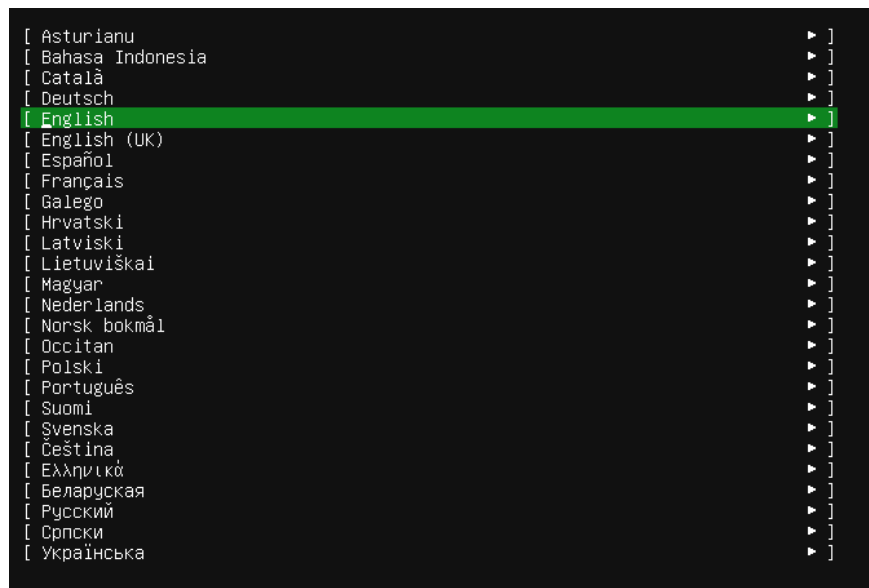


Final virtual machine configuration with a 20 GB disk.

```
[ 22.280456] systemd[1]: Mounted sys-kernel-tracing.mount - Kernel Trace File System.
[ 22.284052] systemd-journald[1067]: Collecting audit messages is disabled.
[ 22.286337] systemd[1]: Finished kmod-static-nodes.service - Create List of Static Device Nodes.
[ OK ] Finished kmod-static-nodes.service..Create List of Static Device Nodes.
[ 22.290864] systemd[1]: Mounted sys-kernel-config.mount - Kernel Configuration File System.
[ OK ] Mounted sys-kernel-config.mount - Kernel Configuration File System.
[ 22.294307] systemd[1]: modprobe@efi_pstore.service: Deactivated successfully
[ 22.296866] systemd[1]: Finished modprobe@efi_pstore.service - Load Kernel Module efi_pstore.
[ OK ] Finished modprobe@efi_pstore.service - Load Kernel Module efi_pstore.
[ 22.303097] systemd[1]: Mounted sys-fs-fuse-connections.mount - FUSE Control File System.
[ OK ] Mounted sys-fs-fuse-connections.mount - FUSE Control File System.
[ 22.311862] systemd[1]: Starting systemd-tmpfiles-setup-dev-early.service - Create Static Device Nodes in /dev gracefully...
Starting systemd-tmpfiles-setup-dev.. Device Nodes in /dev gracefully...
[ 22.363586] systemd[1]: Finished systemd-remount-fs.service - Remount Root and Kernel File Systems.
[ OK ] Finished systemd-remount-fs.service..mount Root and Kernel File Systems.
[ 22.373158] systemd[1]: Starting multipathd.service - Device-Mapper Multipath
```

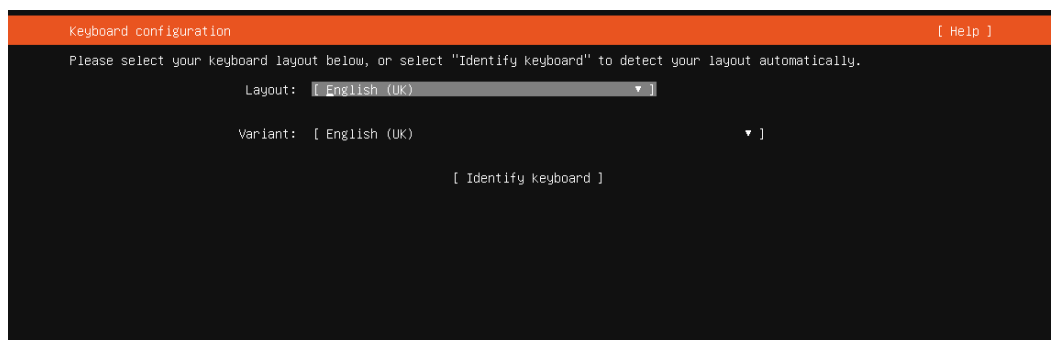
Ubuntu Server installer boot screen after powering on the VM.

Task 5: Choose the Installation Language

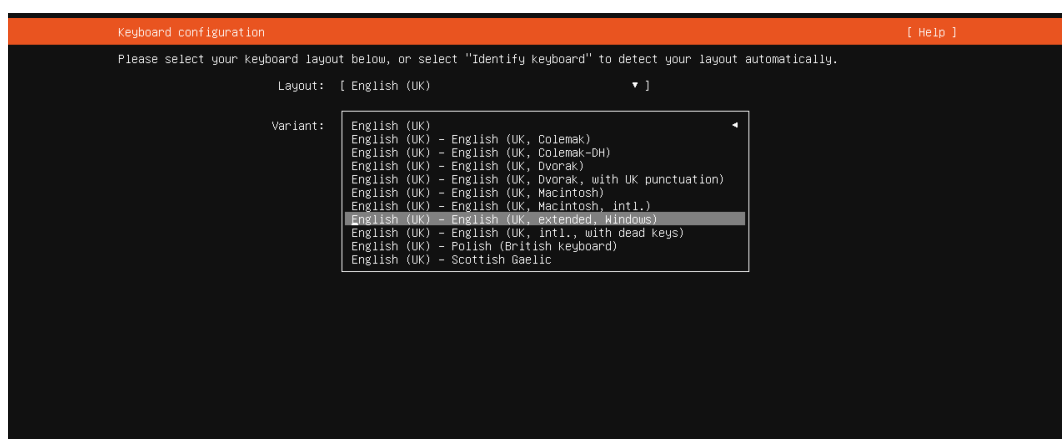


Installation language selected in the Ubuntu installer.

Task 6: Configure the Keyboard Layout

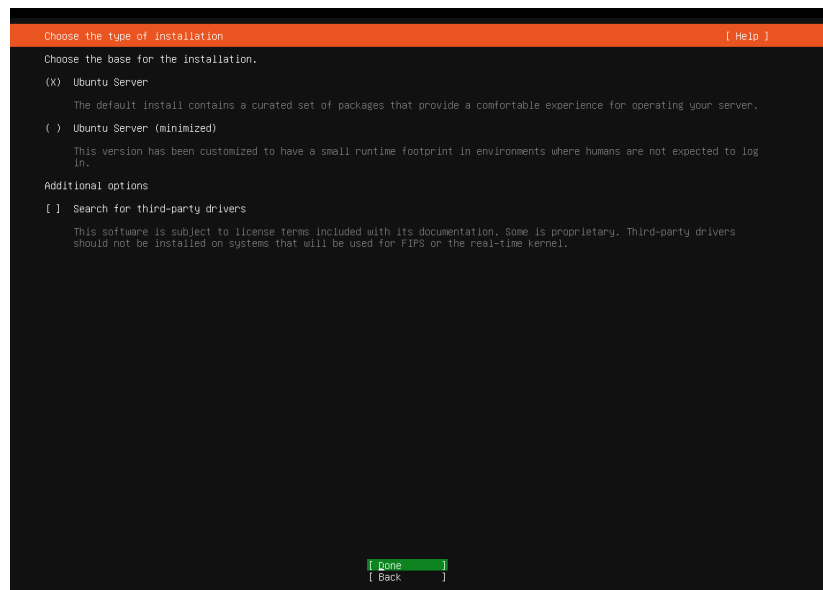


English UK Keyboard layout selected.



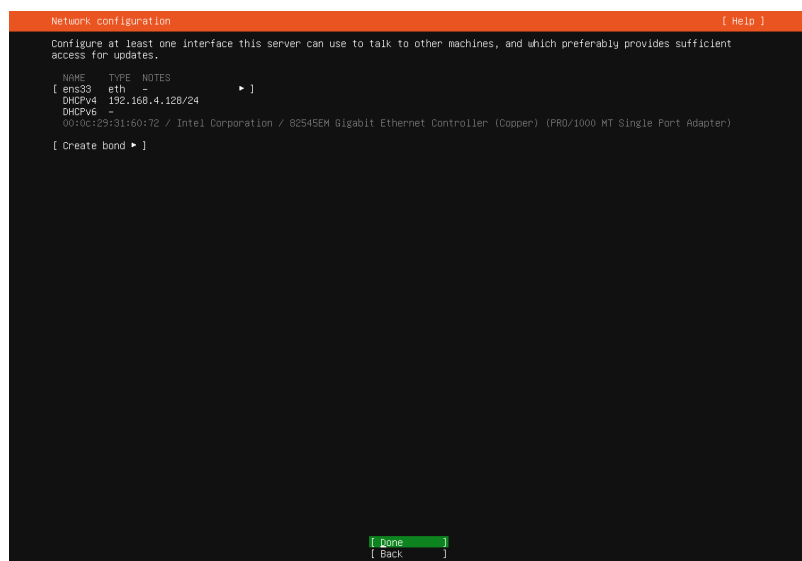
Keyboard variant selected.

Task 7: Choose the Ubuntu Server Installation Type

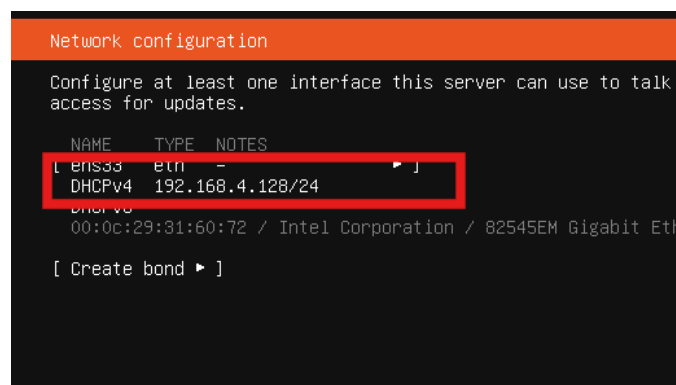


Ubuntu Server installation type selected

Task 8: Configure the Network

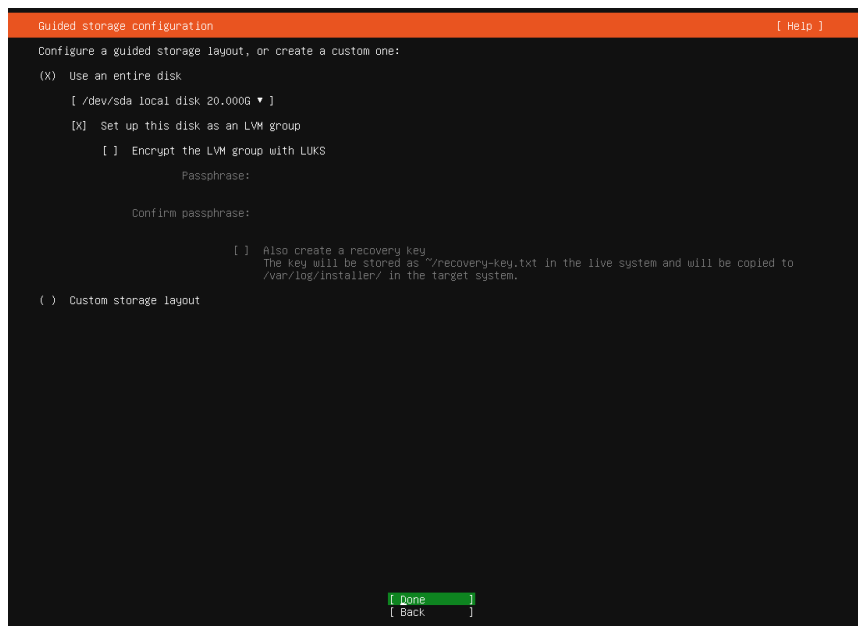


Network interface detected by the installer.

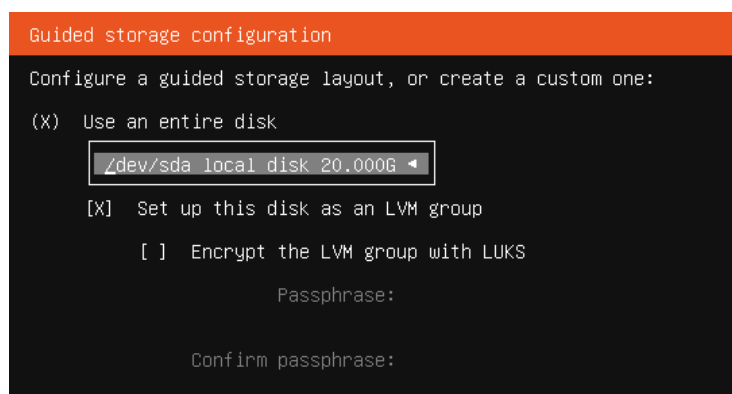


Interface assigned an IP address automatically through DHCP.

Task 9: Configure Storage



Guided storage option selected to use the entire VMware virtual disk.

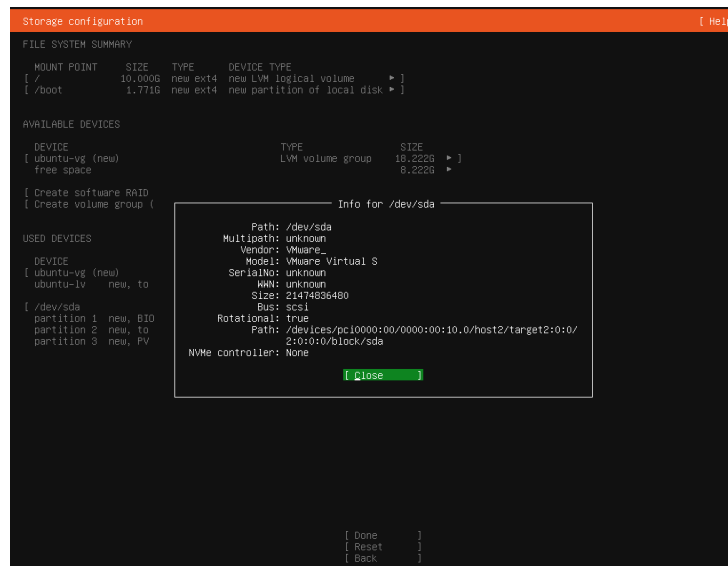


Storage configuration reviewed before continuing.

Task 10: Select the Installation Disk

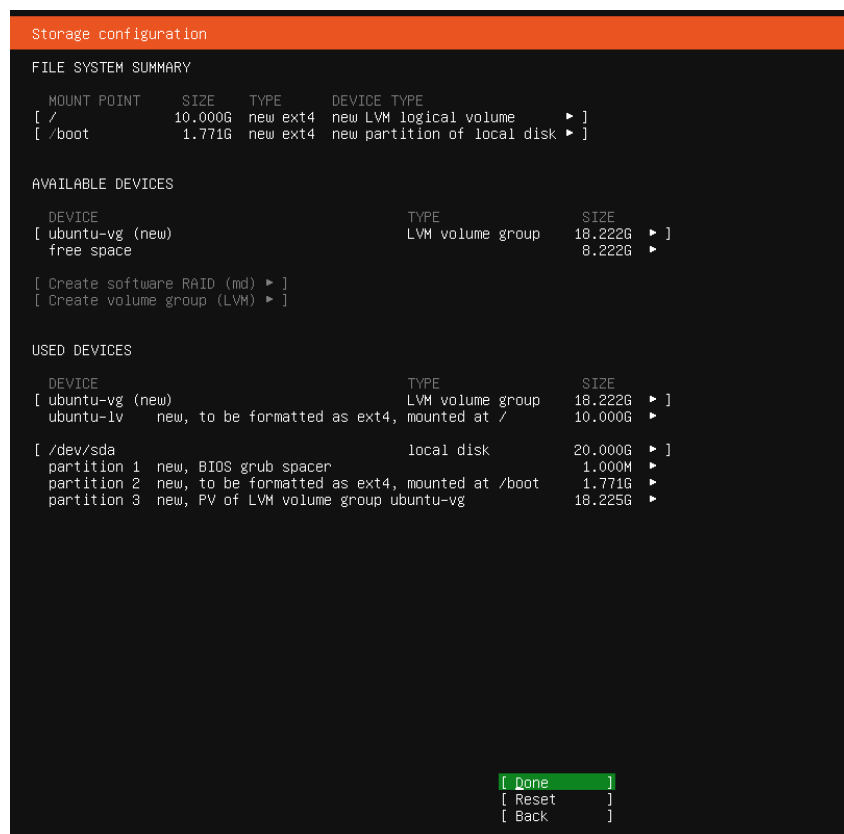
USED DEVICES			
DEVICE		TYPE	SIZE
[ubuntu-vg (new)		LVM volume group	18.222G ▶]
ubuntu-lv	new, to be formatted as ext4, mounted at /		10.000G ▶]
[/dev/sda		local disk	20.000G ▶]
partition 1	new, BIOS grub spacer		1.000M ▶]
partition 2	new, to be formatted as ext4, mounted at /boot		1.771G ▶]
partition 3	new, PV of LVM volume group ubuntu-vg		18.225G ▶]

Installer using /dev/sda, the VMware virtual disk created for this lab.



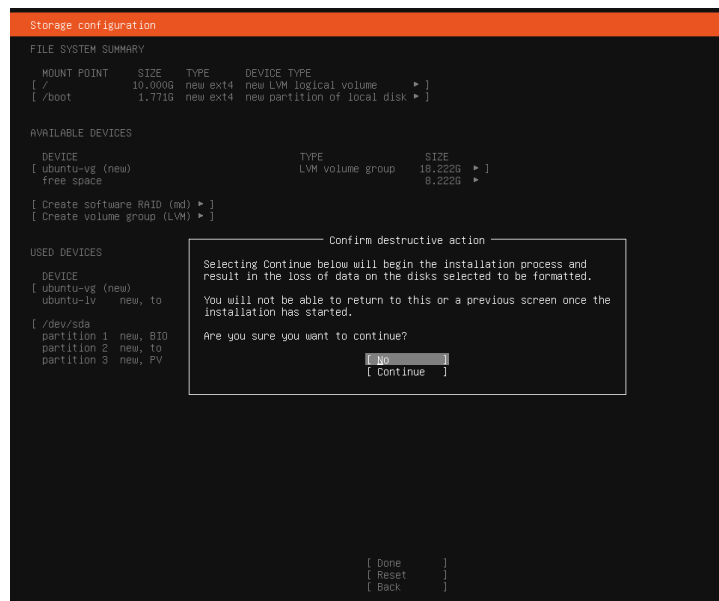
Disk details confirming it is the correct virtual disk.

Task 11: Review the Partitions

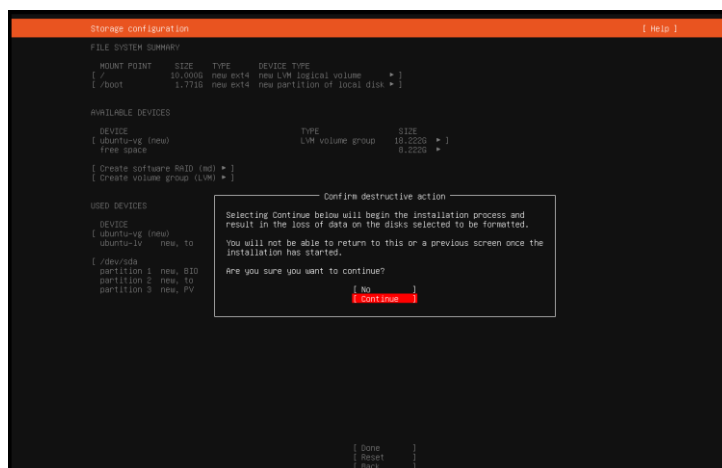


Proposed partition layout on the VMware virtual disk.

Task 12: Confirm the Storage Changes

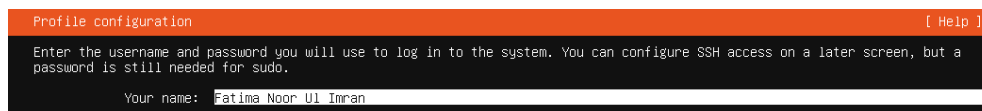


Final destructive-action warning displayed by the installer.

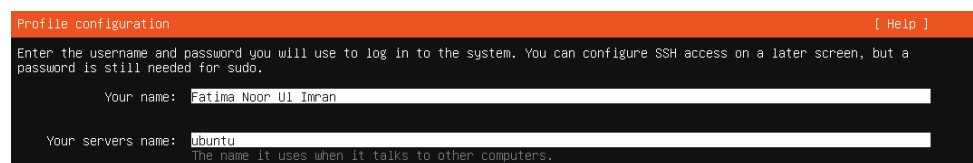


Storage changes confirmed after verifying the correct disk.

Task 13: Configure the Ubuntu Profile



Complete actual name entered in the Your name field.



Server name set to ubuntu.

Profile configuration[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Fatima Noor Ul Imran

Your servers name: ubuntu
The name it uses when it talks to other computers.

Pick a username: fatimanoorulimran

Ubuntu username set to my exact GitHub username, “fatimanoorulimran”.

Profile configuration[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: Fatima Noor Ul Imran

Your servers name: ubuntu
The name it uses when it talks to other computers.

Pick a username: fatimanoorulimran

Choose a password: *****

Confirm your password: *****

[Done]

Completed profile page (password not shown).

Task 14: Complete the Software Installation

Installing system[Help]

```
configuring format: format-0
configuring lvm_voigroup: lvm_voigroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpc_xvxung/mount
configuring Keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring lsscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
curtin command in-target
calculating extra packages to install
installing openssh-server
retrieving openssh-server |
curtin command system-install /
```

[View full log]

Ubuntu Server installation in progress.

Task 15: Reboot and Verify the Ubuntu Identity

```
Installation complete! [ Help ]

curtin command extract
acquiring and extracting image from cp:///tmp/tapc_xxximg/mount
configuring kexec
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NWE over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom

final system configuration
curtin command in-target
calculating extra packages to install
installing openssh-server
retrieving openssh-server
curtin command system-install
unpacking openssh-server
curtin command system-install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquityLateRunBuiltins:
subiquityLateRunUserSupplied:

[ View full log ]
[ Reboot Now ]
```

Installation-complete screen before rebooting.

```
Ubuntu Server X

[ 37.964459] cloud-init[1172]: +---[RSA 3072]-----
[ 37.964672] cloud-init[1172]: |
[ 37.964779] cloud-init[1172]: |. . .
[ 37.964887] cloud-init[1172]: |40 +.
[ 37.964997] cloud-init[1172]: |0. +. .
[ 37.965125] cloud-init[1172]: |+4+....S
[ 37.965235] cloud-init[1172]: |B40+++ .
[ 37.965334] cloud-init[1172]: |+0=B+ .
[ 37.965452] cloud-init[1172]: |.0E.
[ 37.965551] cloud-init[1172]: |. +. +
[ 37.965648] cloud-init[1172]: +---[SHA256]-----+
[ 37.965755] cloud-init[1172]: Generating public/private ecdsa key pair.
[ 37.965869] cloud-init[1172]: Your identification has been saved in /etc/ssh/ssh_host_ecdsa_key
[ 37.965985] cloud-init[1172]: Your public key has been saved in /etc/ssh/ssh_host_ecdsa_key.pub
[ 37.966094] cloud-init[1172]: The key fingerprint is:
[ 37.966300] cloud-init[1172]: SHA256:aw2QjVSabmbSqJ2o29V62aP2N+oq1329f9p1861MGVg root@ubuntu
[ 37.966435] cloud-init[1172]: The key's randomart image is:
[ 37.966534] cloud-init[1172]: +---[ECDSA 256]-----+
[ 37.966629] cloud-init[1172]: |
[ 37.966728] cloud-init[1172]: |. . +
[ 37.967404] cloud-init[1172]: | 0 +.
[ 37.968235] cloud-init[1172]: | 0 +.
[ 37.969223] cloud-init[1172]: | 0 S 0
[ 37.969983] cloud-init[1172]: | B . . .
[ 37.970701] cloud-init[1172]: |. E+..
[ 37.971294] cloud-init[1172]: |. = X+ = +. +0.
[ 37.971378] cloud-init[1172]: |+X48+4000++40
[ 37.973015] cloud-init[1172]: +---[SHA256]-----+
[ 37.973772] cloud-init[1172]: Generating public/private ed25519 key pair.
[ 37.974363] cloud-init[1172]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[ 37.974842] cloud-init[1172]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[ 37.975386] cloud-init[1172]: The key fingerprint is:
[ 37.975841] cloud-init[1172]: SHA256:7y4kv1Sve320hzmigs1xFJMSns+bHI9PC5uAG2VGVs root@ubuntu
[ 37.976365] cloud-init[1172]: The key's randomart image is:
[ 37.976881] cloud-init[1172]: +---[ED25519 256]-----+
[ 37.977384] cloud-init[1172]: |
[ 37.977785] cloud-init[1172]: |.
[ 37.978239] cloud-init[1172]: |..0
[ 37.978644] cloud-init[1172]: |00 . E
[ 37.979003] cloud-init[1172]: | = * * S 0
[ 37.979511] cloud-init[1172]: |.00X . 0.= .
[ 37.979871] cloud-init[1172]: |++80+.00.0.0
[ 37.980306] cloud-init[1172]: |+.+++.0
[ 37.980658] cloud-init[1172]: |..0.+B .
[ 37.981003] cloud-init[1172]: +---[SHA256]-----+

ubuntu login:
ubuntu login:
ubuntu login:
ubuntu login: fatimanoorullinran_
```

Ubuntu login screen after the first boot.

```

ubuntu login:
ubuntu login: fatimanoorulimran
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-34-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Oct  1 03:06:37 AM UTC 2026

System load:  0.51               Processes:    251
Usage of /:   33.8% of 9.75GB    Users logged in: 0
Memory usage: 7%                IPv4 address for ens33: 192.168.4.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

fatimanoorulimran@ubuntu:~$

```

Successful terminal login.

```
fatimanoorulimran@ubuntu:~$ _
```

Prompt fatimanoorulimran@ubuntu:~\$ confirming the required identity.

Task 16: Find the Ubuntu Server IP Address

```

fatimanoorulimran@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:31:60:72 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c29316072
    inet 192.168.4.129/24 metric 100 brd 192.168.4.255 scope global dynamic ens33
        valid_lft 1338sec preferred_lft 1338sec
    inet6 fe80::20c:29ff:fe31:6072/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
fatimanoorulimran@ubuntu:~$ _

```

The ip addr command and its output.

```

fatimanoorulimran@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:31:60:72 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c29316072
    inet 192.168.4.129/24 metric 100 brd 192.168.4.255 scope global dynamic ens33
        valid_lft 1338sec preferred_lft 1338sec
    inet6 fe80::20c:29ff:fe31:6072/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever

```

Server IPv4 address 192.168.4.129 on ens33 with my prompt visible.

Task 17: Connect to Ubuntu Server from Windows

```

PS C:\WINDOWS\system32> ssh fatimanoorulimran@192.168.4.129
The authenticity of host '192.168.4.129 (192.168.4.129)' can't be established.
ED25519 key fingerprint is SHA256:7yAKvLSve3Z0hzmigsixFJMSns+bHI9PC5guAGZvGVs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes

```

SSH command run from Windows.

```
PS C:\WINDOWS\system32> ssh fatimanoorulimran@192.168.4.129
The authenticity of host '192.168.4.129 (192.168.4.129)' can't be established.
ED25519 key fingerprint is SHA256:7yAKvLSve3Z0hzmigsixFJMSns+bHI9PC5guAGZvGVs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.4.129' (ED25519) to the list of known hosts.
fatimanoorulimran@192.168.4.129's password:
```

Server fingerprint accepted by typing yes.

```
fatimanoorulimran@ubuntu: ~$ ssh fatimanoorulimran@192.168.4.129
PS C:\WINDOWS\system32> ssh fatimanoorulimran@192.168.4.129
The authenticity of host '192.168.4.129 (192.168.4.129)' can't be established.
ED25519 key fingerprint is SHA256:7yAKvLSve3Z0hzmigsixFJMSns+bHI9PC5guAGZvGVs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.4.129' (ED25519) to the list of known hosts.
fatimanoorulimran@192.168.4.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-34-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Oct 1 03:24:42 AM UTC 2026

System load:  0.0          Processes:      220
Usage of /:   33.9% of 9.75GB Users logged in: 0
Memory usage: 7%          IPv4 address for ens33: 192.168.4.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

fatimanoorulimran@ubuntu:~$
```

Successful remote SSH login.

```
fatimanoorulimran@ubuntu: ~$ ssh fatimanoorulimran@192.168.4.129
The authenticity of host '192.168.4.129 (192.168.4.129)' can't be established.
ED25519 key fingerprint is SHA256:7yAKvLSve3Z0hzmigsixFJMSns+bHI9PC5guAGZvGVs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.4.129' (ED25519) to the list of known hosts.
fatimanoorulimran@192.168.4.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-34-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
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System information as of Thu Oct 1 03:24:42 AM UTC 2026

System load:  0.0          Processes:      220
Usage of /:   33.9% of 9.75GB Users logged in: 0
Memory usage: 7%          IPv4 address for ens33: 192.168.4.129
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Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

fatimanoorulimran@ubuntu:~$
```

Remote prompt showing fatimanoorulimran@ubuntu.